

Arvin Hedayat

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Intro

Urban infrastructures have always collected data, but what has changed in the past decade is the scale, granularity, and invisibility of that collection. Cities like Montreal now deploy interconnected sensing systems, including surveillance cameras (Ville de Montreal, 2017), automated license plate readers (SPVM, 2024), Bluetooth traffic sensors (Computerworld, 2017), and transit cards that log the movements of residents across public space (Silcoff, 2013). These systems are governed by the Laboratoire d'innovation urbaine de Montreal and marketed under the language of efficiency, safety, and service. Yet the academic literature tells a different story. Scholars argue that such infrastructure constitutes a form of ambient social control, what Bauman and Lyon (2013) call "liquid surveillance," that operates precisely because it is woven invisibly into everyday life.

This paper translates this tension into an empirical question: how do Montreal residents perceive urban sensing technologies, and does their lived experience align with academic frameworks of surveillance as social control? Two sub-questions guide this inquiry: what awareness do residents have of urban sensing infrastructure, and do they frame it as a threat, a public service, or something else, and what shapes that framing? Drawing on semi-structured interviews with nine Montreal residents, this paper finds that the surveillance-as-control framework does not map cleanly onto how residents experience urban sensing in daily life. Awareness is shallow and reactive, and attitudes are shaped less by the technology itself than by trust in the institutions deploying it. These findings demonstrate that public acceptance of urban sensing depends less on awareness than on institutional trust and consent, with

implications for how cities and companies building smart city infrastructure think about legitimacy, transparency, and public accountability.

Literature Review

Urban sensing technologies have expanded rapidly, raising fundamental questions about the relationship between data, infrastructure, governance, and privacy. This section reviews two sets of literature: first, the academic framework of surveillance as social control, and second, empirical work on how the public perceives and accepts these systems.

Surveillance as Social Control

The dominant theoretical framework for understanding urban sensing treats it as an instrument of social control. Bauman and Lyon (2013) argue that surveillance has become “liquid,” no longer confined to fixed institutions but flowing through the infrastructure of everyday life. Unlike the panopticon, in which subjects know they are being watched, liquid surveillance operates through its invisibility, embedding monitoring into systems people use without thinking. Winner (1980) adds that technological artifacts are never politically neutral; a camera at a transit station encodes assumptions about public space and acceptable behavior, reflecting choices about governance that are obscured by the message of efficiency and safety. Brayne (2017) grounds these arguments empirically, showing that data systems like license plate readers extend surveillance reach while rendering it invisible to those being monitored, and that this expansion tends to reproduce existing inequalities. Kitchin (2014) adds that big data is systemically framed as objective and neutral, masking the political choices embedded in its collection. Together, these frameworks suggest urban sensing creates ambient governance that residents are subject to, whether or not they are aware of it.

Public Perception of Urban Sensing

Despite its theoretical strength, the surveillance-as-control framework may overpredict how residents actually experience these systems. Bates et al. (2025) synthesize findings from 50

studies on public space surveillance acceptance. Findings show that attitudes are not black-and-white but conditional, shaped by institutional trust, perceived effectiveness, and the specific technology deployed. The privacy-security tradeoff that dominates academic conversation does not map straightforwardly onto how ordinary people evaluate these systems in practice. Kao and Sapp (2022) offer the most direct empirical support for the institutional trust variable, finding through a nationwide survey that trust in government institutions actively moderates the relationship between privacy concerns and surveillance acceptance. Hassebo (2025) documents the other side of this dynamic, showing that cities consistently frame smart city sensing as a public service, reflecting genuine operational objectives, but also obscuring the amount of data collection from residents who might otherwise object. These studies suggest the gap between academic surveillance theory and lived public experience is a phenomenon worth investigating.

Methods

This paper draws on semi-structured interviews with nine Montreal residents conducted in April 2026. This method was selected because the research question concerns subjective perceptions and lived experience dimensions, which were better captured through open-ended conversation than a structured survey. Participants were recruited through convenience sampling, primarily from the McGill University community. All participant names are pseudonyms to protect anonymity. Interviews lasted approximately 15 to 20 minutes, covering baseline awareness of sensing technologies, reactions to specific infrastructure, attitudes toward the Laboratoire d'innovation urbaine, and broader perceptions of sensing as service or control. Data was analyzed thematically across four dimensions: awareness, framing, institutional trust, and conditionality. The sample is primarily educated young adults in Montreal, limiting generalisability. Findings are exploratory rather than representative of Montreal residents.

Findings

Four themes emerged from analysis: shallow awareness of sensing infrastructure, institutional trust as the primary determinant of attitude, normalisation of control, and possible frameworks centred on consent and transparency.

Awareness Is Shallow and Reactive

Across all nine participants, awareness of urban sensing infrastructure was limited and triggered by visible cues rather than any background understanding of the system. When asked unprompted what technology the city uses to monitor public space, participants consistently named cameras and little else. Automated license plate readers, Opus car data logging, Bluetooth traffic sensors, and pedestrian counters were largely unknown, and the existence of Montreal's Laboratoire d'innovation urbaine was new information for eight of the nine participants.

One interviewee, Jacob, described a pattern common to most: surveillance “doesn’t really cross my mind when I’m in public...the only time I’d really think about it is seeing some sort of camera or a sign that says you’re being recorded.” Another, David, captured a notable contradiction: he “just assumes (he’s) constantly surveilled” yet could not name a single specific technology and was unaware that his Opus card logs every tap. Samuel, a Montrealer who uses public transit regularly, had no knowledge of the city’s coordinated smart city infrastructure at all. His case is notable precisely because of his long familiarity with the city, proving even daily exposure to Montreal’s public infrastructure does not produce awareness of the sensing systems embedded within it. As Brayne (2017) argues, big data surveillance extends its reach precisely because it remains invisible to those being watched.

Institutional Trust as the Determinant of Attitude

The most consistent finding across the dataset was that attitudes were shaped less by the technology itself than by trust in the institutions deploying it. Eight of nine participants expressed different comfort levels depending on whether the city or a private company controlled the data.

The underlying logic was consistent: the government was seen as accountable and bounded by public interest, while private companies were seen as profit-driven and unpredictable in their data use.

David articulated this distinction clearly: “The government is more accountable to people, and I know that there’s a good chance their purpose of the data collection is practical, whereas a private company is doing it for monetary purposes.” Samuel similarly noted that he “generally trusts the government with this kind of stuff,” but that private involvement would “definitely change things.” The Toronto Google case (Carter and Rieti, 2020), introduced as a factual prompt, crystallised this divide in nearly every interview, producing discomfort even among participants who had expressed general comfort with urban sensing.

Caleb stands as an outlier, inverting the typical preference by suggesting that if a company has his data, “it might as well be a private company so they do productive stuff with it.” His reasoning reflects confidence in private sector innovation rather than accountability concerns, illustrating that public-private trust is not universal but shaped by prior familiarizations toward government and markets. This pattern mirrors the quantitative findings of Kao and Sapp (2022), who demonstrate that institutional trust actively moderates the relationship between privacy and surveillance acceptance.

Normalization of Control and the Consent Problem

When presented with the academic framing that urban sensing amounts to social control, five participants said it resonated, and four found it overstated. On top of that, no one said the conversation would change how they moved through the city. Noah agreed the framing “100% resonates,” explaining that “having cameras is meant to control, it’s meant to keep you in line societally,” yet remained comfortable, describing himself as used to it from growing up in a place where surveillance is prevalent. Jacob accepted a “good sort of surveillance” as a necessary condition of urban life. People can know they are subject to a system of control and feel entirely at ease within it.

The sharpest critique came from Joseph, who reframed the academic argument directly: “I don’t even think people don’t realize they’ve consented to it. I think they just haven’t consented to it. It’s not like they send out referendums. It’s just kind of imposed.” This shifts the problem from personal awareness to structural consent. A fourth theme reinforced this: without prompting, participants articulated conditions under which sensing would feel legitimate. Transparency and public accountability recurred as such criteria. Thomas offered an observation regarding this: “Even apps we have that give us terms and conditions are kind of digital consent, but for some reason cameras and government surveillance don’t want to apply that.” This identifies a structural inconsistency in how consent operates across data-collecting objects, and aligns with the governance standards Hassebo (2025) identifies as necessary for responsible smart city infrastructure.

Discussion

The findings complicate the surveillance-as-control framework. First, Bauman and Lyon (2013) argue that liquid surveillance is structurally ambient, diffused through everyday infrastructure rather than concentrated in fixed institutions. The findings here add a qualitative dimension that their framework overlooked. Participants could not name the technologies monitoring them, did not know the city’s smart city office existed, and had never considered that their transit card logs their movements. David assumed constant surveillance while being unable to identify a single technology doing the surveilling, demonstrating a cultural belief, rather than a felt experience grounded in actual knowledge. The Montreal case highlights Bauman and Lyon’s argument: liquid surveillance flows through everyday life while remaining opaque to the people it monitors. Second, the institutional trust finding also extends Winner’s (1980) argument that artifacts have politics in their design. Participants evaluated urban sensing in relation to who held the data. A license plate reader felt acceptable under city control and concerning under Google’s. This suggests that the politics of an artifact are not fully encoded in its design but are also constituted

through the social relationships surrounding its deployment. Kitchin's (2014) point about data being framed as neutral is supported here, but what residents pushed back against was not neutrality, but institutional accountability. They did not ask whether the data was objective; they asked who was responsible for it.

Third, Joseph's distinction between unawareness and non-consent is the most theoretically significant finding in the data set. The surveillance-as-control literature tends to frame the problem as one of false knowing, with residents unknowingly subject to systems they would resist if they knew. Joseph's formulation rejects this; the problem is not ignorance, but the absence of a consent mechanism. Thomas' parallel between an app's terms of service and the absent equivalent for city sensing reinforces the point. Residents have developed expectations for data consent through their experience of digital platforms, and they apply those expectations to urban infrastructure even when no mechanism exists to satisfy them.

Taken together, the findings suggest that the break between surveillance theory and lived experience is not a matter of indifference. Residents are largely unaware of the specific infrastructure monitoring them, inclined to trust the institutions operating it, and given little mechanism to consent to or contest it. The gap between the academic critique of surveillance and how it is actually lived is not a failure of awareness but a product of invisibility, distrust, and absent accountability structures. As Hassebo (2025) argues, responsible smart city governance requires transparency about what data is collected, by whom, and for what purpose. The residents in this study arrived at the same conclusion independently, suggesting the legitimacy gap in urban sensing is not a matter of public education but of structural accountability.

Conclusion

This paper set out to ask whether Montreal residents experience urban sensing in ways that align with academic theories of surveillance. The answer is largely no, but for reasons that refine rather than refute the theory. Awareness of specific infrastructures is shallow, attitudes are

governed by institutional trust more than technology type, and the felt experience of control is largely absent even among participants who intellectually accept that control is the system's primary function. The surveillance-as-control framework captures something real. What it misses is the role of invisibility, institutional trust, and the absence of consent mechanisms in shaping lived experience.

For cities and companies using smart city infrastructure, these findings suggest that public acceptance is not primarily a communication problem. Residents are not hostile to urban sensing because they misunderstand it; they are uncomfortable when they cannot identify who is accountable, when data use seems opaque or commercially motivated, and when no framework exists for their participation in the decision.

Future research should examine these questions across more diverse populations and city contexts, as this sample was primarily young, educated residents. Older residents, those in peripheral neighborhoods, and communities with higher levels of police presence would likely tell another story. What remains is this: cities and other organizations are building sensing infrastructure faster than they are building the public legitimacy to sustain it. Whether that gap can be closed through better governance, greater transparency, or new frameworks for public consent remains a question worth asking.

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